

CHAPTER 1: HUMAN LANGUAGE AND LANGUAGE SCIENCE

In this chapter, we begin to explore what language is, and how language scientists (also known as linguists) think about it and observe it. It might well be that most of experience learning about language has had to do with rules that you could get right or wrong. That's not the approach we're taking in this book. Instead, we're going to look at how to use the tools and techniques of linguistics to observe the patterns of human languages. From these observations, we'll try to draw some conclusions about the abstract principles and organization of human language in people's minds and in language communities. Along the way, we'll also consider the ways that language science and people's attitudes about language have bolstered colonial structures of power and privilege and have been used to do harm.

When you've completed this chapter, you'll be able to:

- Differentiate between prescriptive and descriptive ways of thinking about language,
- Identify components of mental grammar,
- Explain some properties of all human languages,
- Describe some techniques for doing language science, and
- Discuss the ethics of doing language science.

1.1 WHAT EVEN IS LANGUAGE?



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We're all users of language. Right now I'm talking and writing to you in a variety of Canadian English, and I bet many of you also know one or more other languages. Linguistics is the scientific study of human language. That definition is short, but it's not exactly simple, is it? How do we study language scientifically? And what even is language?

The word *language* is used for several different complex concepts that are interconnected with each other. One use of the word is to refer to individual languages, like American Sign Language (ASL), Basque, English, Langue des signes québécoise (LSQ), Nishnaabemwin, Xhosa, and many others.

The word *language* can also refer to other, related notions. If you're a programmer, you might have a section on your résumé that lists the computer languages you know, like Python, R, C++, or Perl. Computer languages are not usually the focus of linguistics, even though many linguists use them to analyse linguistic data! There are also metaphorical uses of the word *language*, such as body language or love languages. These uses of the word are also outside of what linguistics usually studies.

For the moment, let's think about one particular language, because it happens to be the one we're using now. I'm using one variety of English (a variety spoken by a middle-aged white lady in Ontario). Before I started making this video, I used my fingers to type words on my keyboard. Now as I read those words and talk to you, I'm squeezing the air out of my lungs; I'm vibrating my vocal folds, and I'm manipulating parts of my mouth to produce sounds. Those sounds are getting captured by a microphone and recorded on my computer, then I'll upload them to the eBook. If you're listening to this video, the sounds I recorded are playing on your device,

and your eardrums are reacting to the auditory information. If you're reading the text or the captions, your eyes are reacting to the visual information. Your eyes and your ears send signals to your brain. And somehow, after all that, if my communication was successful, you end up with an idea in your mind that's similar to the idea in mine. There must be something that we have in common to allow that to happen: some shared system that allows us to understand each other's ideas through language. This shared system is what many linguists call the **mental grammar**, and one of the goals of linguistics is to find out what that shared system is like.

So we've focused our definition of linguistics a little bit, by saying that we're interested in the scientific study of human language, of the grammar, the shared system that allows us to understand each other. What is the grammar like? Or to put it another way, what do we know when we know a language?

What is grammar?

Imagine you're an alien, you've just arrived on Earth, and you need to figure out how to understand the language used in the particular earthling community that you've landed in. What kinds of things do you need to figure out? One of the first things you'll need to know about that language is what counts as talking. Is this language signed or vocalized? In other words, what is the **modality** of the language? Many human languages are **vocalized** (or "spoken"). In this modality, language users make sounds with their larynx, tongue, teeth and lips, and receive sounds with their ears. Other human languages are **signed**. Language users make signs with their fingers, hands, wrists and forearms, and receive signs by sight or by touch. Even though they have very different modalities, sign and vocal languages share many properties in their grammars. In this book, we'll try to reserve the words *speaking* and *speech* for vocal languages, and refer to **language users** when we're talking about languages of any modality. In other places you might also see *linguaging* used as a verb to mean "using language in any modality".

Once you've figured out the modality, what next? You probably need to segment the stream of auditory or visual information into meaningful units. By observing carefully, you might be able to figure out that a particular sequence of sounds or gestures recurs in this language, and that some consistent meaning is associated with that sequence. For example, maybe you've noticed that the language users you've encountered make the sounds "cookie" as they're offering you a round, sweet, delicious baked good. Or maybe you've noticed that when that word has a *z* sound at the end of it, *cookies*, you're being offered more than one of them!

The part of the grammar that links up these forms with meanings is the **mental lexicon**. It's a bit like a dictionary in your mind. Knowing a word in a language involves recognizing its form – the combination of signs or sounds or written symbols, and its meaning. For the majority of words in the world's languages, the link between form and meaning is **arbitrary**.



Figure 1.1. Cookies.



Figure 1.2. Pumpkin.

For example, the English word for this thing is *pumpkin* and the Nishnaabemwin word is *kosmaan*. There's nothing inherently orange or round or vegetabley about either of those word forms: the pairing of that meaning to that form is arbitrary in each language. (But there are words whose form has an iconic, less arbitrary relationship their meaning; we talk about them more later in this book.)

Suppose you've figured out that cookies are delicious and you want to ask your earthling hosts for more of them. To do that, you need to figure out how to control the muscles of your mouth, tongue, and lips to speak the word for cookie, or how to use your hands, fingers, wrists and forearms to sign the word. In other words, you need to know something about the **articulatory phonetics** of the language. This brings up an important point about grammar: when we know a language fluently, a lot of our grammatical knowledge is unconscious,

or **implicit**. For the languages that you know, your knowledge of the lexicon is probably fairly conscious or explicit, and probably also some of your knowledge about your language's **morphology**: that's the combinations of meaningful pieces inside words (like how if you want more than one cookie you say *cookies* with a *z*). But you're probably not as conscious of things like how you use your articulators to make the sounds *k* or *z*.

Our implicit knowledge of language also includes **phonology**, information about how the physical units of language can be combined and how they change in different contexts. **Syntax** is the part of your mental grammar that knows how words can or can't be combined to make phrases and sentences, much of which is implicit. Syntax works hand in hand with **semantics** to allow the grammar to calculate the meanings of these phrases. And the **pragmatics** part of the mental grammar can help you to know what meanings arise in different contexts. For example, "I have some news," could be interpreted as good news or bad news depending on the context.

All of these things are parts of the grammar: the things we know when we know a language. But a lot of this knowledge is implicit, and the thing about implicit knowledge is that it's hard to observe. One of the most important jobs we're doing in this textbook is trying to be explicit about what mental grammar is like, and about what kinds of evidence we can use to figure that out. We'll talk about this challenge more in [Section 1.3](#) below.

What about reading and writing?

I bet you're wondering why I didn't include reading and writing as part of the mental grammar above. After all, as a student you probably invested a lot of time into learning how to read and write. And those skills are indeed part of the grammatical knowledge you have about your language. But language users don't actually need to know how to read and write to have a mental grammar. It's common for kids in Canada to start learning to read and write around age five, but they are pretty competent in the phonetics, phonology, morphology, syntax and semantics of one or more languages before they ever go to school.

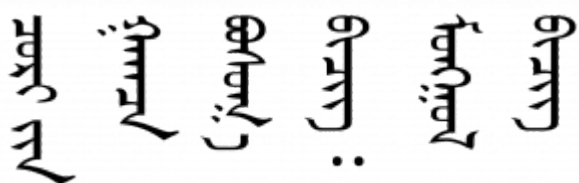


Figure 1.3a. Vertical Mongolian Script.

Монгол

Figure 1.3b. Cyrillic Mongolian script.

Furthermore, language users could start using a different writing system without changing anything else about the grammar. Mongolian, for example, presently uses two different writing systems: the Cyrillic alphabet and traditional Mongolian script, which is written vertically. Speakers of Mongolian understand each other's speech no matter which script they use to record the language in writing. And there are plenty of human languages that just don't have written forms. Signed languages like ASL and LSQ, for example, don't have written forms. Most signers are bilingual in their sign language and in the written form of another language.

So, because not every human language has a reading and writing system and not every language user has access

to reading and writing systems, we consider these skills to be secondary parts of the mental grammar. If you're literate in your language, then that literacy is certainly woven into your mental grammar. But literacy isn't necessary for grammatical competence.

Check your understanding



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1.2 WHAT GRAMMARS ARE AND AREN'T



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The previous section was a very quick tour of some of the parts of the mental grammar. We'll be discovering a lot more about grammar throughout this book. Notice that we're using the term *grammar* a little differently from how you might have encountered it before. Maybe your experience of grammar is as a textbook or style guide with a set of rules in it, rules that lead to consequences if you break them — you'll lose points on your essay or get corrected with a red pen. What we're most interested in this book is the mental grammar: the system in your mind that allows you to understand and be understood by others who know your language. Every human language has a mental grammar: that's how the users of each language understand each other!

This is a really important idea. One way that people sometimes express racist, colonialist and ableist ideas is to deny the validity of a language by claiming that it “has no grammar”. But the truth is that **all languages have grammar**. All languages have a system for forming words, a way of organizing words into sentences, a systematic way of assigning meanings. Even languages that don't have alphabets or dictionaries or published books of rules have users who understand each other; that means they have a shared system, a shared mental grammar. Using linguists' techniques for making scientific observations about language, we can study these grammars.

The other important thing to keep in mind is that no grammar is better than any others. Maybe you've heard someone say, “Oh, I don't speak real Italian, just a dialect,” implying that the dialect is not as good as so-called real Italian. Or maybe you've heard someone say that Québec French is just sloppy; it's not as good as the French they speak in France. Or maybe you've heard someone say that nobody in Newfoundland can speak proper English, or nobody in Texas speaks proper English, or maybe even nobody in North America speaks proper English and the only good English is the Queen's English that they speak in England. From a linguist's

point of view, **all languages and dialects are equally valid!** There's no linguistic way to say that one grammar is better or worse than another. This is part of what it means to study grammar from a scientific approach: scientists don't rate or rank the things they study. Ichthyologists don't rank fish to say which species is more correct at being a fish, and astronomers don't argue over which galaxy is more posh. In the same way, doing linguistics does not involve assigning a value to any language or variety or dialect. We also need to acknowledge, though, that many people, including linguists, do attribute value to particular dialects or varieties, and use social judgments about language to create and reinforce hierarchies of power, privilege and status. We'll look at some examples in [Section 1.4](#) and in [Chapter 2](#) we'll go into more detail about the terms *language*, *variety*, and *dialect*.

One of the most fundamental properties of grammar is **creativity**. One obvious sense of the word *creative* has to do with artistic creativity, and it's true that we can use language to create beautiful works of literature. But that's not the only way that human language is creative. The sense of creativity that we're most interested in in this book is better known as **productivity** or **generativity**. Every language can create an infinite number of possible new words and sentences. Every language has a finite set of words in its vocabulary – maybe a very large set, but still finite. And every language has a small, finite set of principles for combining those words. But every language can use that finite vocabulary and that finite set of principles to produce an infinite number of sentences, new sentences every single day.

A consequence of the fact that grammar is productive is that **languages are always changing**. Have you heard your teachers or your parents say something like, “Kids these days are ruining English! They should learn to speak properly!” Or if you grew up speaking Mandarin, maybe you heard the same thing, “Those teenagers are ruining Mandarin! They should learn to speak properly!” For as long as there has been language, older people have complained that younger people are ruining it. Some countries, like France and Germany, even have official institutes that make rules about what words and sentence structures are allowed in the language and which ones are forbidden. But the truth is every language changes over time. Languages are used by humans, and as humans grow and change, and as our society changes, our language changes along with it. Some language change is easy to observe in the lexicon: we need to introduce new words for new concepts and new inventions. For example, the verb *google* didn't exist when I was an undergraduate student, and now googling is something I do nearly every day. Languages also change in their phonetics and phonology, and in their syntax, morphology and semantics. In [Chapter 10](#) we'll look at the systematic ways that linguists study variation and change.

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1.3 STUDYING LANGUAGE SCIENTIFICALLY



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We said that linguistics is the science of human language. When we say that linguistics is a science, that doesn't mean you need a lab coat and a microscope to do linguistics. Instead, what it means is that the way we ask questions to learn about language uses a scientific approach.

The scientific way of thinking about language involves making systematic, **empirical observations**. That word *empirical* means that we observe data to find the evidence for our theories. All scientists make empirical observations. Entomologists observe the life cycles and habitats of insects. Chemists observe how substances interact. Linguists observe how people use their language. Just like entomologists and chemists, linguists aim for an accurate description of the phenomenon they're studying. And like other scientists, linguists strive to make observations that are not value judgments. If an entomologist observes that a certain species of beetle eats leaves, she's not going to judge that the beetles are eating wrong, and tell them that they'd be more successful in life if only they ate the same thing as ants. Ideally, the same would be true of linguists — we wouldn't go around telling people how they should or shouldn't use language. Of course, like all scientists, and like all humans, linguists have biases that often prevent us from reaching this ideal; more on this later in the book. But the goal for doing language science is to do so with a **descriptive** approach to language, not a **prescriptive** approach, to describe what people do with their language, but not to prescribe how they should or shouldn't do it.

For example, you could describe English plurals this way:

Adding *-s* to a noun allows it to refer to many of something, like *apples*, *books*, or *shoes*.

Or you could prescribe how you think people should form plurals this way:

Because the word *virus* is derived from Latin, you should pluralize it as *viri*, not *viruses*.¹

So when we're doing linguistics, our goal is to make descriptive, empirical observations of language. But one challenge to being a language scientist is that a lot of what you're studying is hard to observe. Unlike our entomologist friends, we can't just go out to the garden and poke around and find some grammar crawling on a plant. We have to figure out how to make observations about the mind. Throughout this book you'll get introduced to the many different tools of language science, which allow us to make systematic observations of how humans use language.

Going meta: Observing what's possible in a language

As I keep saying, a lot of the linguistic knowledge we have is unconscious. One of the tools we can use to get at our mental grammar is to try to access **metalinguistic awareness**, that is, the conscious knowledge you have about your grammar, not the grammatical knowledge itself. If you've studied a language in school you probably have some metalinguistic awareness about it because you got taught it explicitly. But for your first language, the one you grew up speaking, it can be a little more difficult to access your metalinguistic knowledge because so much of it is implicit. It's a skill that we'll keep practicing throughout this book.

Here's an example of accessing your metalinguistic awareness. Say you want to create a new English word for a character in a game. Are you going to call your cute little creature a *blifter* or a *lbitfer*? Neither of those forms exists in English, but they both use sounds that are part of English phonetics. You probably have a strong feeling that *blifter* is an okay name for your new creature, while *lbitfer* is a pretty terrible name. Notice that your sense that *lbitfer* is wrong is not a prescriptive sense — it's not that it sounds rude or you'll get in trouble for combining those sounds that way. It just ... can't happen. You've made a descriptive observation that *lbitfer* is not a possible word in English. From that observation, we can conclude that *lbitfer* is ungrammatical in English.

Since linguistics uses the word *grammar* in a particular way, the words *grammatical* and *ungrammatical* also have a specific meaning. An **ungrammatical** word or phrase or sentence is something that just can't exist in a particular language: the mental grammar of that language does not generate it. Notice that grammaticality isn't about what actually exists in a language; it's about whether a form could exist. In this example, both *blifter* and *lbitfer* have the same sounds in them, but *blifter* could be an English word and *lbitfer* couldn't. In other words, *blifter* is grammatical in English and *lbitfer* is ungrammatical in English.

1. This prescriptive statement doesn't reflect what really happens in English, since most English speakers talk about *viruses*, not *viri*. And in fact, it doesn't even reflect what happens in Latin, since the Latin word *virus* did not have a plural form!

It's often useful to compare similar words, phrases or sentences to try to access our metalinguistic awareness. Let's look at another example of observing what's possible. Here are two similar sentences, both of which are possible (or acceptable) in English.

- a. Sam compared the forged painting with the original.
- b. Sam compared the forged painting and the original.

Let's try to make questions out of these sentences:

- c. Did Sam compare the forged painting with the original?
- d. Did Sam compare the forged painting and the original?

Observing those two questions, we can see that both (c) and (d) are acceptable in English. Now let's try a different kind of question:

- e. What did Sam compare the forged painting with?
- f. *What did Sam compare the forged painting and?

Comparing these two sentences gives us a really clear finding: (e) is possible, but (f) is not. We use an asterisk or star at the beginning of sentence (f) to indicate that it just can't happen. These **acceptability judgments** (also sometimes known as **grammaticality judgments**) are our empirical observations: these two similar sentences are both possible as declarative statements (a-b) and as yes-no questions (c-d), but when we try to make a wh-question out of them, the result is acceptable for the first one (e) but not for the second one (f). Having made that observation, now our job is to figure out what's going on in the mental grammar that can account for this observation. Why is (e) grammatical but (f) isn't?

More tools for language science

Because it can be tricky to access metalinguistic knowledge, you might not want to rely on the acceptability judgments of one single language user. Instead, you could use a survey to gather quantitative data about acceptability from many users. We can also use **surveys** to **elicit** the words that people use for particular items. From survey data we know that some people call this thing a sweatshirt, other people call it a hoodie, and people in Saskatchewan call it a bunny hug. Surveys are particularly useful for learning about regional variation, which you can learn more about in [Chapter 10](#). If you're studying regional and social variation you might also gather data using **interviews**, in which you could ask questions like, “Does the ‘u’ in *student* sound like the ‘oo’ in *too* or the ‘u’ in *use*?”.



Figure 1.4. Hoodie.

A **corpus** is another tool that allows us to make language observations. A corpus is a big database that collects examples of language as used in the world, from books, newspapers, message boards, videos. Some corpora contain only written text, and others include video of signed language, or audio files with phonetic transcription. The nice thing about tools like acceptability judgments, surveys, and corpora is that they're relatively easy to use: you don't need a lot of training or money to ask people what word they use for athletic shoes, or to see how a word or phrase is used in a corpus. We'll use some of these accessible tools throughout this book.

There are also more specialized tools for doing language science. Phoneticians use a variety of software for analyzing audio and video recordings of speakers and signers. Praat (Boersma & Weenink, 2022) is a popular waveform editor for analyzing audio recordings. While Praat is specialized for linguists, it has some similarities to audio-editing programs for podcasting. ELAN (*ELAN | The Language Archive*, 2021) is a powerful tool that allows a user to annotate video recordings, and the program SLP-Annotator (Lo & Hall, 2019) also enables phonetic **annotations** of video-recorded sign language. Some phoneticians also make anatomical measurements of the articulators, using ultrasound or palatography for speech or motion capture for signing.

We can draw on techniques from behavioural psychology to make observations about language use in real-time using **experiments**. You might measure reaction times and reading times for words and sentences, or ask participants to listen to words that are mixed with white noise. Some experiments use eye-tracking to measure people's eye movements while reading a text, watching a signer, or listening to a speaker. It's even possible to use **neural imaging** techniques like electroencephalography (EEG) and functional magnetic resonance

imaging (fMRI) to observe brain activity during language processing. You can learn more about interpreting the data from these experiments in [Chapter 13](#).

When you're starting out in linguistics, it's often really exciting to use the scientific method to think about grammar, as you start to see that grammar is not just a set of arbitrary rules to memorize so you sound "proper". Even if we're not peering through a microscope wearing a lab coat, the tools of language science allow us to make systematic observations of how humans use language. And we can interpret those observations to draw conclusions about the human mind.

Check your understanding



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1.4 THINKING ABOUT STANDARDS AND “PROPER” GRAMMAR



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In previous sections we learned that one of the goals of doing linguistics is to describe languages and dialects accurately without ranking any dialect as better than any other. This is actually a pretty radical goal, because of course language is a deeply human behaviour, and therefore is deeply intertwined with human relationships and social categories. Relationships like teacher-student, doctor-patient, or customer-server, for example, all involve power relations that play a role in people’s expectations about language. Likewise, the communities that we belong to, whether they’re based on ethnicity, religion, profession, fandom, or any other social category, shape how we use language and how we expect others to use language. So when we’re studying language scientifically, we can’t separate the grammar from all the other social pieces.

So we’re striving for this radical goal of considering **all languages and dialects as equally valid** from a linguistic point of view, but we also have to acknowledge that people have attitudes and expectations that arise from social power dynamics, and these attitudes – whether positive and negative – lead to linguistic **bias**. Everyone, including linguists, has linguistic biases. We can’t help making judgments about people based on how they use language. But by learning to think about the relationship between language and power, we can gain **metalinguistic awareness** of our own linguistic biases, at the same time as we’re developing metalinguistic awareness of our grammars.

Here’s an example of a linguistic bias that’s really prevalent in Canada and the US. North Americans tend to perceive all the varieties of British English as having **high prestige**. They tend to assume that speakers of UK English are better educated and more intelligent than speakers of North American varieties of English — even for varieties that have low prestige in the United Kingdom. Someone who speaks a variety that’s stigmatized

in the UK might arrive in Canada to find that everyone thinks their English is very fancy. Their English hasn’t changed, but people’s attitudes towards it have!

Language Standards and “Standard” Languages

Some ways of using language are associated with higher prestige. Because of colonialism, these are often the forms of language used by white people, by wealthier people, or by people who have received more formal education.

When people talk about the “standard” variety of a language, they usually mean the form that has been **standardized**, that is, the form that most closely matches the language used in dictionaries, textbooks, and high-status media. This standardization happens via social mechanisms of power. In France, for example, there’s an official government body, the *Académie Française*, that decides what counts as correct, standard French. In 2017, when they noticed more and more French writers including feminine nouns and adjectives alongside the standard masculine forms, they published a declaration that this kind of inclusive writing was a mortal danger (“*un péril mortel*”) for French! It’s their literal job to tell people they’re languaging wrong.

Unlike French, English does not have an official language police to enforce prescriptive language rules, but that doesn’t mean the standardized varieties of English are any less connected to power and privilege. Instead, standardized English is enforced through social norms, through dictionaries and style guides, textbooks and grammar-checking software. There’s no official Boss of Canadian English warning about the dangers of gender-inclusive language, but it was still a big deal when the in-house style guide of the *Globe and Mail*, a national newspaper in Canada, decided in 2017 that it was okay to use specific singular *they*. And in the UK, the shorthand term for the highest-prestige variety is “the Queen’s English” — who has more power and privilege than a monarch?

To be clear, the “standard” that these authorities enforce isn’t chosen out of nowhere, and is not somehow objectively determined to be the best or clearest variety. (Remember there’s no linguistic way to determine “best” when it comes to language.) The standard is usually just the variety that’s associated with economic, social, or political power. For many languages, the “standard” is whatever variety is spoken in the capital city, or by a dominant political class. For English and for other European languages, the variety that people categorize as “standard” tends to be the variety that white people with a certain amount of formal education use.

In North America, we can observe how assumptions about standard languages intersect with our ideas of race by considering the variety of English associated with Black speakers. You’ll see this variety called **African American Language (AAL)**, **African American Vernacular English (AAVE)**, **Black English**, or **Ebonics**, depending on who’s talking about it and when. I’ll use the term *Black English* here, following Calhoun et al. (2021), to include people who use this variety and who are part of the African diaspora but not necessarily

African-American. One important thing to note is that not all Black folks in the United States speak Black English, and not all speakers of Black English are Black, but using this variety of English is a strong indicator of Black American identity. Even though Black English is characteristically American, and has many speakers across many different regions of the USA, it's somehow not what anyone means when they refer to Standard American English. That reveals the common linguistic bias: when people say "standard language" they usually think "the language that white people use".

Isn't it good to have standards?

You might think of having a standardized variety of a language as a good thing, or at least as a neutral thing. We're used to having a single variety of English appear in most written sources, for example. It's easy to view standardization as positive if the variety that you and your family used when you were growing up was relatively close to the standardized variety used in schools. But if we assume that the standardized form is the only correct or proper form, we end up discriminating against users of different varieties. Here are some examples:

- More than 90% of people Haiti speak Kreyòl, a language with its own consistent grammar and spelling. But public education in Haiti is offered in standardized French. So when kids start school, they get told by their teachers that their language is wrong (Degraff & Stump, 2018). The same pattern holds true for kids who speak Black English in most US schools. It's harder for them to learn!
- A judge in Alberta disregarded the medical evidence provided by an expert witness, a doctor who spoken Nigerian English. In his ruling, the judge made it clear that he distrusted the doctor's medical opinion because his accent was not Canadian. (Grant, 2019)
- A Black deaf man who signed in Black ASL was imprisoned in an institution for decades because the signers who assessed him categorized his variety of ASL as incoherent, so they labelled him as languageless and incompetent. (Burch & Joyner, 2007)

We'll learn about more examples in Chapter 2.

Because elementary and high schools usually teach language in a prescriptive way, you've probably internalized the assumption that the standardized variety of your language is the best or most correct variety, and maybe even the assumption that languages have to have standards. Using your growing metalinguistic awareness, you can start to question why some varieties are considered standard and others aren't. It's likely that the answers to those questions have more to do with social status than with grammar.

Check your understanding



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CHAPTER 5: MORPHOLOGY

In this chapter, we look at words and at the meaningful pieces that combine to create words. We will see that languages vary in how words are built, but that nonetheless we can find **structure** inside of words in all languages. In linguistics, the study of word forms is known as **morphology**.

When you've completed this chapter, you'll be able to:

- Identify morphologically complex words, and the morphemes within them
- Distinguish between inflectional morphology, derivational morphology, and compounds
- Explain how a word's morphology interacts with its lexical category
- Analyze the structure of complex words

5.1 WHAT IS MORPHOLOGY?



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In linguistics, **morphology** is the study of how words are put together. For example, the word *cats* is put together from two pieces: *cat*, which refers to a particular type of furry four-legged animal (🐱), and *-s*, which indicates that there's more than one such animal (🐱🐱🐱).

Most words in English have only one or two pieces in them, but some technical words can have many more, like *non-renewability*, which has at least five (*non-*, *re-*, *new*, *-abil*, and *-ity*). In many languages, though, words are often made up of many parts, and a single word can express a meaning that would require a whole sentence in English.

For example, in the Harvaqtuurmiutut variety of Inuktitut, the word *iglujjualiulaqtuq* has 5 pieces, and expresses a meaning that we can translate into English with the sentence “They (sg) made a big house.” (*iglu* = house, *-jjua* = big, *-liu* = make, *-lauq* = distant past, *-tuq* = declarative; this example is from a 2010 paper by Richard Compton and Christine Pittman).

Not all combinations of pieces are possible, however. To go back to the simple example of *cat* and *-s*, in English we can't put those two pieces in the opposite order and still get the same meaning—*scat* is a word in English, but it doesn't mean “more than one cat”, and it doesn't have the pieces *cat* and *-s* in it, instead it's an entirely different word.

One of the things we know when we know a language is how to create new words out of existing pieces, and how to understand new words that other people use—at least, as long as those new words are made of pieces we've encountered before. We also know what combinations of pieces are not possible. In this chapter we'll

learn about the different ways that human languages can build words, as well as about the structure that can be found inside words.

What is a word?

If morphology is the investigation of how words are put together, we first need a working definition of what a **word** is.

In everyday life, in English we might think of a word as something that's written with spaces on either side. This is an **orthographic** (or spelling-based) definition of what a word is. But just as writing isn't necessarily a reliable guide to a language's phonetics or phonology, it doesn't always identify words in the sense that is relevant for linguistics. And not all languages are written with spaces in the way English is—not all languages have a standard written form at all. So we need a definition of “word” that doesn't rely on writing.

The definition of “word” is actually a hotly debated topic in linguistics! Linguists might distinguish **phonological words** (words for the purposes of sound patterns), **morphological words** (words for the purposes of morphology), and **syntactic words** (words for the purposes of sentence structure), and might sometimes disagree about the boundaries between some of these.

For the purposes of linguistic investigation of grammar we can say that a word is the **smallest separable unit in language**.

What this means is that a word is the smallest unit that can stand on its own in an utterance. For example, content words in English (**nouns, verbs, adjectives, and adverbs**) can stand by themselves as one-word answers to questions, as you can see in the mini-dialogues in (1).

- (1) a. What do you like to eat?

Answer: cake (noun)

- b. What did you do last night?

Answer: sleep (verb)

- c. What colour is the sky today?

Answer: orange (adjective)

- d. How did you wake up this morning?

Answer: slowly (adverb)

Words are also **syntactically independent**, which means they can appear in different positions in a sentence,

changing their order with respect to other elements even while the order of elements inside each word stays the same.

Even though words are the smallest *separable* units of language, that doesn't mean that words are the smallest unit of language overall. As we already saw earlier in this section, words themselves can have smaller pieces inside them, as in the simple cases of *cats* (*cat-s*) or *non-renewability* (*non-re-new-abil-ity*)—but these smaller pieces can't stand on their own.

To refer to these smaller pieces within words, we use the technical term **morpheme**. A morpheme is the smallest systematic pairing of both form (sign or sound) and meaning or grammatical function. (We say “meaning or grammatical function” instead of just “meaning” because while some morphemes have clear meanings, of the type that will be discussed in [Chapter 7](#) in the context of lexical semantics, other morphemes express more abstract grammatical information.)

Words that contain more than one morpheme, like *cats* or *nonrenewability* or *iglujjualiulaqtuq*, are **morphologically complex**. Words with only a single morpheme, like *cat* or *new*, are **morphologically simple**.

Ask yourself if the word “morphology” itself is morphologically complex. Can you identify morphemes within this word, systematic pairs of form and meaning? Historically, this word is built from two morphemes borrowed from Classical Greek: *morph-* “shape” and *-ology* “study of”. People who know English don't necessarily know Classical Greek, though. Regardless of a word's etymology (the history of a word), the question of whether it is morphologically complex is a question about how people who know that word use it today. A word might be morphologically complex for some people, but morphologically simple for others. Neither of those options is “correct” or “incorrect”, they just represent different grammars in the minds of language users.

In linguistics *morphology* is the study of word shapes. In biology, though, *morphology* is the study of the shape of animals and other organisms, and if you do an internet search for “morphology”, the first hits often relate to the biological meaning.

Our goal in morphology is to understand how words can be built out of morphemes in a given language. In the rest of this chapter we will first look at the shapes of different morphemes (and morphological processes); in later sections we will review different functions that morphology can have, looking at divisions between **derivational** morphology, **inflectional** morphology, and **compounding**.

Check your understanding



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5.2 ROOTS, BASES, AND AFFIXES



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Affixes vs roots

Morphemes can be of different types, and can come in different shapes. Some morphemes are **affixes**: they can't stand on their own, and have to attach to something. The morphemes *-s* (in *cats*) and *inter-* and *-al* (in *international*) are all affixes.

The thing an affix attaches to is called a **base**. Just like whole words, some bases are morphologically simple, while others are morphologically complex.

For example, consider the word *librarian*. This word is formed by attaching the affix *-ian* to the base *library*.

Librarian can then itself be the base for another affix: for example, the word *librarianship*, the state or role of being a librarian, is formed by attaching the affix *-ship* to the base *librarian*.

There is a special name for simple bases: **root**. A root is the smallest possible base, which cannot be divided, what we might think of as the *core* of a word. Roots in English we've seen so far in this chapter include *cat*, *library*, and *nation*.

We can ignore the fact that 'y' turns into an 'i' when a suffix attaches to *library*. There's a convention of English spelling that 'y' becomes 'i' before an affix; it doesn't reflect any change in phonological shape.

If you look at the history of the words *library* and *nation*, they both trace back to Latin (by way

of French), and in Latin the relevant words were morphologically complex: *library* traces back to the Latin root *libr-* (meaning “book”), and *nation* traces back to the Latin root *nat-* (meaning “be born”). When a child first encounters a word like *library* or *nation*, however, the word doesn’t come annotated with this historical information! In the minds of most contemporary English speakers, it is likely that *library* and *nation* are treated as simple roots; in [Chapter 13](#), you’ll learn about how this kind of hypothesis could be tested experimentally.

Turning back to affixes, an affix is any morpheme that needs to attach to a base. We use the term “affix” when we want to refer to all of these together, but we often specify what *type* of affix we’re talking about.

A **prefix** is an affix that attaches **before** its base, like *inter-* in *international*.

A **suffix** is an affix that **follows** its base, like *-s* in *cats*.

A **circumfix** is an affix that attaches **around** its base.

An **infix** is an affix that attaches **inside** its base.

A **simultaneous affix** is an affix that takes place at the **same time as** its base.

Prefixes and suffixes are very common, not only in English but also in other languages. Circumfixes, infixes, and simultaneous affixes are less common, and so we’ll look at examples of each in order.

Circumfix

An example of a **circumfix** can be found in the marking of **plural possessors** in many Algonquian languages. The following examples are from Meskwaki, spoken in parts of the Midwest of the US and in Northern Mexico; the source of these examples is Oxford (2020), who adapted them from an in-preparation grammar by Amy Dahlstrom (*A grammar of Meskwaki, an Algonquian language*). These examples are presented in Meskwaki orthography; “a.” indicates a long vowel.

- (2) a. ne-ta-nes-aki
 1-daughter-AN.PL
 “my daughters”
- (2) b. ne-ta-nes-ena-n-aki
 1-daughter-1PL-AN.PL
 “our daughters”

What you can see here is that the singular possessor in “my daughters” is marked only by a prefix, but the plural

possessor in “our daughters” is marked by the combination of the prefix *ne-* and the suffix *-ena·n*—or, in other words, by a **circumfix**.

These examples have **morpheme-by-morpheme glosses**, which means that the morphological analysis has been done for you; in [Section 5.11](#) we’ll discuss how we figure out the boundaries between morphemes in a language we aren’t already familiar with.

Glossed examples include at least three lines: the first line gives the example in the original language, usually in either a phonetic transcription or the language’s own orthography. The second line gives the meaning or function of each word or each morpheme (if the words are divided into morphemes). The third line gives a translation of the whole example into the language the author is writing in, which in this textbook is English.

Morpheme-by-morpheme glosses use standard abbreviations:

- **1** stands for “first person” (I, me, my / we, us, our)
- **PL** stands for “plural” (so **1PL** means “we, us, our”)
- **AN** stands for “animate”. Algonquian languages distinguish all nouns as “animate” or “inanimate”, and this is reflected in its morphology.

Infix

Infixes are affixes that appear **in the middle of another morpheme**. For example, in Tagalog (a language with about 24 million speakers, most of them in the Philippines) the infix *-um-* appears immediately after the first consonant of the base to which it attaches. This infix expresses perfective aspect for verbs. Perfective aspect indicates completed action, usually translated with the English simple past:

- (3) a. [takbuh] *run* [tumakbuh] *ran*
 b. [lakad] *walk* [lumakad] *walked*
 c. [bili] *buy* [bumili] *bought*
 d. [kain] *eat* [kumain] *ate*

For an affix to be an infix, it must appear inside another morpheme, not just in the middle of a word. If you look at the word *unluckiness* (*un-luck-y-ness*), for example, *-y* is a suffix that just happens to appear in the middle

of the word because another suffix (*-ness*) attaches after it. But *-y* still isn't an infix, because it attaches after its base (*luck*), not inside its base.

Simultaneous affix

Simultaneous affixes are common in signed languages and in languages with tone. When signing, it's possible to do things with multiple articulators (a second hand, or your face), or to add motion on top of a sign, in a way that is not possible with oral articulations in spoken languages.

For example, in ASL there is a morpheme that attaches to verbs to express **continuative aspect** (the meaning that something happens continuously for a while, or for a long time). This morpheme involves adding a particular circular motion to the base sign for the verb; this circular motion doesn't happen before or after the verb, but simultaneously with it. You can see the application of this affix in the first and last videos for the verb *STUDY* [in this linked article](#) from the online Handspeak ASL dictionary (Lapiak 1995–2022) (the second video in that post shows the application of a different simultaneous affix, one for iterative aspect).

There is morphology in some spoken languages that has a similar profile. For example, languages with tone sometimes have tonal morphemes, where a change in tone expresses grammatical information, while the consonants and vowels of the base stay the same.

English isn't a tonal language, but we have some pairs of words that clearly involve the same root, but where the stress has shifted. These are noun-verb pairs where the noun has stress on the first syllable, but the verb has stress on the second syllable.

- (4) a. They used to use **réc**ords to **recó**rd music.
- b. I have a **pér**mit that **permít**s me to drive.
- c. I receive mail at my home **ád**dress, at least when it's **addréssed** properly.

Not all English speakers have stress shift in the same pairs of words. For example, while I pronounce *address* with stress on the first syllable when it's a noun, many people pronounce it with stress on the second syllable (*addréss*) for both the noun and the verb.

Free and bound morphemes

Another way to divide morphemes is by whether they are **free** or **bound**. A **free** morpheme is one that can occur as a word on its own. For example, *cat* is a free morpheme. A **bound** morpheme, by contrast, can only occur in words if it's accompanied by one or more other morphemes.

Because affixes by definition need to attach to a base, only **roots** can be free. In English most roots are free, but we do have a few roots that can't occur on their own. For example, the root *-whelmed*, which occurs in *overwhelmed* and *underwhelmed*, can't occur on its own as **whelmed*.

In many other languages, though, all (or most) roots are **bound**, because they always have to occur with at least some morphology. This is the case for verbs in French and the other Romance languages, for example; it was also the case for Latin, which is why the roots *nat-* and *libr-* were shown with hyphens above.

We show that morphemes are bound by putting hyphens either before or after them, on the side that they attach to other morphemes. This applies to bound roots as well as to affixes.

Check your understanding



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CHAPTER 6: SYNTAX

In [Chapter 5](#) we looked at the internal structure of words (morphology). In this chapter we look at how words are organized into **phrases** and **sentences**, which in linguistics is called **syntax**.

When you've completed this chapter, you'll be able to:

- Use the evidence of constituency tests to identify the phrases within a sentence
- Categorize words into lexical and functional categories based on their distribution
- Identify relationships between grammatically related sentences (active and passive, statements and questions)
- Draw tree diagrams to represent the structural analysis of sentences in English

Alternative paths through this chapter

The first half of this chapter ([6.1](#) to [6.12](#)) introduces core concepts and argumentation in syntax, while tree diagrams and X-bar theory are covered in the second half ([6.13](#) to [6.21](#)). This structure has been chosen to emphasize core concepts over the details of tree structure.

Many introductory courses in linguistics instead interleave core concepts and their formalization in tree diagrams, however. For such courses—or for independent learners who prefer to follow such a structure—we recommend reading sections in the following order:

- Basics
 - [6.1 Syntactic knowledge and grammaticality judgements](#)
 - [6.2 Word order](#)
 - [6.3 Phrases, Heads, and Selection](#)
 - [6.4 Identifying phrases: Constituency tests](#)
 - [6.5 Functional categories](#)

- Introducing trees
 - [6.13 From constituency to tree diagrams](#)
 - [6.14 Trees: Introducing X-bar theory](#)
 - [6.15 Trees: Sentences as TPs](#)
 - [6.16 Trees: Modifiers as adjuncts](#)
 - [6.17 Trees: Structural ambiguity](#)
- Complex sentences
 - [6.6 Clausal embedding](#)
 - [6.18 Trees: Embedded clauses](#)
- Syntax of questions
 - [6.7 Main clause Yes-No questions](#)
 - [6.8 Main clause content questions](#)
 - [6.9 Embedded content questions.](#)
 - [6.19 Trees: Movement](#)
- Argument structure
 - [6.10 Arguments and thematic roles](#)
 - [6.11 Changing argument structure: Causatives and passives](#)
 - [6.20 Trees: Movement beyond questions](#)
 - [6.21 Trees: Summary](#)
- (This path through the chapter skips [6.12 Interim summary](#))

Each section ends with a “Navigation” section with links that can be used to follow this interleaved path.

Appendix 1 introduces trees using phrase structure rules instead of X-bar theory. Courses adopting that formalization should continue with [Appendix 1](#) after [6.13 From constituency to tree diagrams](#).

6.1 SYNTACTIC KNOWLEDGE AND GRAMMATICALITY JUDGEMENTS



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In linguistics, **syntax** is the study of how words are organized into phrases and sentences. Just as the morphemes in a word are organized into structures, the words in a sentence are also best viewed not just as a string of words, but instead as having a **hierarchical structure**. And just as words contain a head morpheme, we'll see that every phrase has an element that is its **syntactic head**.

What kind of knowledge do we have about the syntax of a language we know? Let's start by considering the sentence in (1):

- (1) All grypnos are tichek.

You might not know what a *grypno* is, or what it means to be *tichek* (because these are made-up words!), but you can tell that this sentence is still the right kind of “shape” for English. In other words, (1) is consistent with the way English speakers put words together into sentences.

Compare this with the sentence in (2):

- (2) *Grypnos tichek all are.

Unlike (1), (2) isn't the right shape for a sentence in English. Even if you did know what a *grypno* was, or what it meant to be *tichek*, this still wouldn't be the way to put those words together into a sentence in English.

Something we can be pretty confident about is that you've never heard or read either of these sentences before encountering them in this chapter. In fact, most of the sentences you encounter in this textbook are likely to

be made up of words you haven't heard or read in exactly that order before. So that means that your internal grammar of English must be able to generalize to new cases—this is the **generativity** of language, something introduced back in [Chapter 1.2](#).

As someone who uses language—in the case of (1) and (2), as someone who speaks and reads English—you can identify sentences that do or do not fit the patterns required by your internal grammar. In syntax we describe sentences that do match those patterns as **grammatical** for a given language user, and sentences that do not match required patterns as **ungrammatical**.

Grammaticality judgements in syntax

In syntax when we say something is **ungrammatical** we don't mean that it's “bad grammar” in the sense that it doesn't follow the type of grammatical rules you might have learned in school. Instead, we call things ungrammatical when they are inconsistent with the grammatical system of language user.

The evaluation of a sentence by a language user is called a **grammaticality judgement**. Grammaticality judgements are a tool for investigating the linguistic system of an individual language user—there is no way to get a grammaticality judgement for “English” as a whole, only grammaticality judgements from individual English speakers. Despite that, sometimes you will see a sentence described as grammatical or ungrammatical “in English” or another language; technically this is a shorthand for saying that users of English (or another language) generally agree about whether it is grammatical or not. In many cases different users of a language disagree about the status of a particular example, and this tells us something about **syntactic variation** in that language!

We are often most interested in examples that are **ungrammatical**, because they tell us about the limits on building sentences in a language. The convention in linguistics is to mark ungrammatical examples with an **asterisk** (*) at the beginning of the sentence, sometimes called a **star** (which is slightly easier to say). Whenever you see that symbol in front of an example in this chapter, as in example (2) above, it indicates that the example is ungrammatical in the linguistic sense.

Sometimes we want to indicate that a sentence is weird because of its meaning, rather than its syntax. In these cases we use a hashmark symbol (#) instead of a star.

For example, consider an example like (3):

- (3) #The book pedalled the ladder harmoniously.

This sentence is the right *shape* for English, it just doesn't make any sense—books can't pedal anything, ladders

can't be pedalled, and it's not clear how you could pedal something harmoniously. So we would say that the sentence in (3) is grammatical but semantically odd, and that's what the hashtag symbol indicates.

Most of the sentences we will consider in this chapter are ones that many English speakers (but not all) share similar judgements about. If you disagree with any of the judgements reported here, you can take the opportunity to think about what that tells you about your own grammar, and whether the difference could be explained using the tools we develop here. If the theory developed here can't account for your pattern of grammaticality judgements, that shows that we need to make changes to the theory!

The goals of syntactic theory

Our goal in syntax is to develop a theory that can do two things:

1. Predict which sentences are grammatical and which ones are ungrammatical
and
2. Explain the observed properties of grammatical sentences.

In much of this chapter we'll focus on the syntax of varieties of English, because English is a language that's common to everyone who reads this textbook. At the same time, we will often have opportunities to see how other languages show us the scope of variation for syntax in human languages. In the end, we want to build a theory that can be used to explain not just properties of English, but properties of *all* human languages.

What kind of theory do we need to make these kinds of predictions? If languages were **finite** we could simply list all the grammatical sentences, state that anything not on the list was an ungrammatical sentence, and be done. But any language user can generate sentences that no one has ever encountered before, and other users of the same language can understand those sentences, so what we “know” when we know the syntax of a language must be more than just a list of grammatical sentences. In the next section of this chapter, we'll see that what we know about syntax can't be just about the order of words, it has to be something about their grouping (**constituency**) as well.

Check your understanding





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Navigation

If you are following the alternative path through this chapter that interleaves core concepts with tree structures, you should continue with [6.2 Word order](#).

6.4 IDENTIFYING PHRASES: CONSTITUENCY TESTS

Video Part 1:



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Video Part 2:



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By identifying certain parts of sentences as **phrases**, we are making a claim that language users represent them as units in their mental grammar. The technical term for units inside a sentence is **constituent**: a constituent is any group of words that acts together within a sentence.

Along with headedness, constituency is one of the central concepts in syntax. Both of these are highlighted when we represent the structure of language using **tree diagrams**, as we'll see beginning in [Section 6.13](#), but they're fundamental to understanding the organization of sentences with or without trees.

When we analyze a new sentence, how do we identify the phrases inside of it? We want to find evidence that certain groups of words actually do act together as units. To find that evidence, we use grammaticality judgements, and a few simple tests.

The tests that identify constituents (often called **constituency tests**) that we'll review in this chapter come in four basic types:

- Replacement tests
- Movement tests
- It-clefts
- Answers to questions

Many textbooks also introduce a **coordination** test, but it is not always reliable, so we'll discuss it briefly at the end of this section but won't rely on it.

REPLACEMENT TESTS

Here are two sentences to start with.

(1) The students saw a movie after class.

(2) The students saw a movie about dinosaurs.

Let's consider the string of words *a movie*. Based on discussion so far in this chapter, you might have the idea that this is a noun phrase—or at least that it *could* be a noun phrase. But whether or not you have that idea, we need evidence to decide one way or the other.

One piece of evidence that something is a noun phrase is that you can **replace** it with a pronoun, and get a sentence *with the same meaning* (in a context where the meaning of the pronoun is made clear). In (3) we take the pronoun *it* and replace the string of words we're interested in, then ask if the new sentence is grammatical and whether it has the same meaning.

(3) The students saw a movie after class. → The students saw **it** after class.

Replacing *a movie* with *it* in (3) does give us a new grammatical sentence that can mean the same thing as (1), so we have evidence not only that *a movie* is a constituent in (1), but also that that constituent is a noun phrase.

What about *a movie* in (2)? Let's run the same test there:

(4) The students saw a movie about dinosaurs. → *The students saw **it** about dinosaurs.

This time the result of replacing *a movie* with *it* is an ungrammatical sentence, so in (2) *a movie* is not a complete noun phrase. We might be surprised about this—we expect a noun like *movie* to be inside a noun phrase—but if we test other possible constituents we see that it's not that there's no noun phrase here, it's just that the noun phrase is a bit bigger. As shown in (5), it turns out that we can replace *a movie about dinosaurs* with *it* and get a grammatical sentence.

(5) The students saw a movie about dinosaurs. → The students saw **it**.

Based on comparing the results of our replacement tests in (4) and (5), we can conclude that in (2) *a movie* is not a complete noun phrase, but *a movie about dinosaurs* is both a constituent and a noun phrase.

We can do the same pronoun replacement test with the string *the students* in (1). Because *students* is plural, the relevant pronoun is *they*:

(6) The students saw a movie after class. → **They** saw a movie after class.

The result of this replacement is grammatical, so we conclude that *the students* is also a constituent, and also a noun phrase.

Replacement tests don't have to involve pronouns. Verb phrases can be replaced with *do* (or *do too*), but seeing this usually requires setting up two sentences with different subjects or with a contrast in time like *yesterday* vs. *today*. Since we have just seen that *the students* in (1) is a noun phrase subject (because it comes at the beginning of a simple declarative sentence, before the verb), let's set up a replacement test for verb phrase with a preceding sentence with a different subject:

(7) a. The teachers saw a movie after class, and... → The students **did too**.
 b. The teachers saw a movie after class, and... → *The students **did too** before class.

What we see in (7) is that *did too* can replace *saw a movie after class*, but can't replace *saw a movie* alone. This tells us that *saw a movie after class* is a constituent, and it's a verb phrase (because *do (too)* replaces verb phrases).

What about the string *after class*? This string expresses a time, and we can replace it with the word *then*:

(8) The students saw a movie after class. → The students saw a movie **then**.

This shows that *after class* is a constituent; in fact, it's a **prepositional phrase**. Not all prepositional phrases can be replaced by *then*, however—*about dinosaurs* is also a prepositional phrase, but can't be replaced by *then*.

(8) The students saw a movie about dinosaurs. → *The students saw a movie **then**.

Here the result of doing replacement would be grammatical in other contexts, but it isn't another way to say that the students saw a movie about dinosaurs—this is why it's marked ungrammatical here, it's ungrammatical *on the intended meaning*. **You have to pay attention to both grammaticality and meaning when you do replacement tests.**

At this point, you're probably wondering how you know what you can use as a replacement when running these tests. Here are some handy tips that will work for most English speakers:

- Noun Phrases can be replaced with pronouns (*it, them, they*).
- Verb Phrases can be replaced with *do* or *do too* (or *did, does, doing*).
- Some Prepositional Phrases (but not all) can be replaced with *then* or *there*.
- Adjective Phrases can be replaced with something that you know to be an adjective, such as *happy* (though in this case the meaning will change)

Because replacement is category-specific, you can use the evidence of replacement tests both to identify constituents *and* to figure out the constituent's category: If you can replace it with a pronoun, then you've got a noun phrase and you can look for the noun that's the head. If you can replace it with *do (too)*, then you've got a verb phrase which will have a verb as its head.

MOVEMENT TESTS

Replacement is not the only tool we have for checking if a set of words is a constituent. Some constituents can be moved to somewhere else in the sentence without changing the sentence's meaning or its grammaticality. Prepositional phrases are especially good at being moved. Consider this sentence:

(9) Nimra bought a scarf at that strange little shop.

Let's start by targeting the last string of words by moving it to the beginning. Move the string of words then ask yourself whether the resulting sentence is grammatical.

(10) Nimra bought a scarf at that strange little shop. → **At that strange little shop** Nimra bought a scarf.

It is! In isolation the sentence might sound a little unnatural, but we can imagine a context where it would be

fine, such as, “At the department store she bought socks, at the pharmacy she bought some toothpaste, and at that strange little shop, she bought a scarf.”

On the other hand, if we target a smaller string of words, as in (11), we get a different result.

- (11) Nimra bought a scarf at that strange little shop. → ***At that strange** Nimra bought a scarf little shop.

The result of moving the string *at that strange* to the beginning of the sentence is a total disaster. The fact that the resulting sentence is totally ungrammatical gives us evidence that the string of words *at that strange* is **not** a constituent in this sentence.

CLEFT TEST

A **cleft** construction is one where you take two parts of a sentence and divide them from each other. (A *cleft* is a split or gap.)

In English, a cleft is a sentence with the form: *It is/was _ that _*.

To use the cleft test, we take the string of words that we’re investigating and put it after the words *It was* (or *it is/it’s*), then put the remaining parts of the sentence after the word *that*. Let’s try this for phrases that we’ve already shown to be constituents with our other tests.

- (12) The students saw a movie after class.
→ It was **a movie** that *the students saw _ after class*.
→ It was **after class** that *the students saw a movie _*.

- (13) The students saw a movie about dinosaurs.
→ It was **a movie about dinosaurs** that *the students saw _*.

- (14) Nimra bought a scarf at that strange little shop.
→ It was **at that strange little shop** that *Nimra bought a scarf _*.

To cleft a verb phrase in English you need put a present or past tense form of *do* in the position the verb phrase occupied in the original sentence, as shown in (15).

- (15) The students saw a movie after class.
 → (?)It was **see a movie after class** that the students *did*.

Clefting a verb phrase doesn't always sound totally natural in English, but most people find it better than clefting non-constituents. At the end of this section, go back and compare (15) to the sentences in (16) and (17), and see if you agree that (15) is better.

By contrast, things that our tests showed were *not* constituents cannot be put into the first position of a cleft sentence:

- (16) *It was **a movie** that *the students saw* _ *about dinosaurs*.
 (17) *It was **at the strange** that *Nimra bought a scarf* _ *little shop*.

Now let's try the cleft test on a new sentence:

- (18) Rathna's brother baked these delicious cookies.
 → It was **these delicious cookies** that *Rathna's brother baked* _.
 → It was **Rathna's brother** that _ *baked these delicious cookies*.

The cleft test shows us that the string of words *these delicious cookies* is a constituent, and that the words *Rathna's brother* are a constituent. But look what happens if we apply the cleft test to other strings of words:

- (19) Rathna's brother baked these delicious cookies.
 → *It was **Rathna's brother baked** that _ *these delicious cookies*.
 (20) Rathna's brother baked these delicious cookies.
 → *It was **these delicious** that *Rathna's brother baked* _ *cookies*.
 (21) Rathna's brother baked these delicious cookies.
 → *It was **cookies** that *Rathna's brother baked these delicious* _.

All of these applications of the cleft test result in totally ungrammatical sentences. For (19) and (20) this gives

us evidence that those underlined strings of words are not constituents in this sentence. In (21), though, what we're testing is a single word, and single words are always constituents—they always act together as a unit. So what (21) shows is that *cookies* by itself is not a complete phrase.

Remember, though, just because a certain string of words isn't a constituent in one sentence, doesn't mean it's not a constituent in *any* sentence—the result of a constituency test only applies to the specific sentence you're testing.

ANSWERS TO QUESTIONS

If a string of words is a constituent, it's usually grammatical for it to stand alone as the answer to a question based on the sentence. We can see this in the sentences in (22).

- (22) Rathna's brother baked these delicious cookies.
- a. What did Rathna's brother bake? **These delicious cookies.**
 - b. Who baked these delicious cookies? **Rathna's brother.**

Answers to questions are also a good context for *do*-replacement (as a **replacement test** to identify verb phrases):

- (23) Who baked these delicious cookies? Rathna's brother **did**.

In the answer, "Rathna's brother did", the word *did* replaces the verb phrase *baked these delicious cookies*.

Again, if a string of words is not a constituent, then it is unlikely to be grammatical as the answer to a question. In fact, it's difficult to even form the right kind of question:

- (24) a. *What did Rathna's brother bake cookies? *These delicious.
 b. *Who of Rathna's these delicious cookies? *Brother baked.

COORDINATION TEST

In linguistics, **coordination** refers to joining two elements together with a word like *and* or *but* (these words are called **coordinators**). Coordination is also called **conjunction**; the elements you join together are called **conjuncts**.

In most cases, each conjunct is a constituent. For this reason, many textbooks introduce coordination as a

constituency test. For example, it's possible to coordinate a noun phrase like *these delicious cookies* with a noun like *dessert*, which helps show that *these delicious cookies* is a noun phrase. It's also possible to coordinate a verb plus its object (like *baked these delicious cookies*) with an intransitive verb (like *left*), which helps show that *baked these delicious cookies* is a verb phrase.

- (25) a. Rathna's brother baked [these delicious cookies] and [dessert].
 b. Rathna's brother [baked these delicious cookies] and [left].

The problem with coordination as a constituency test is that there are a few constructions, both in English and other languages, where the conjuncts are not constituents. In a construction called **Right Node Raising**, for example, two coordinated sentences or verb phrases seem to share an object that comes after both of them:

- (26) Rathna's brother baked _ and Rathna herself ate _ these delicious cookies

You might think that because they're connected with *and*, [*Rathna's brother baked*] and [*Rathna herself ate*] are constituents, but other constituency tests will show that they're not—and after all, if the verb plus its object is always a constituent, then these strings can't be! For this reason, if you use coordination as a constituency test, it's always a good idea to make sure that at least one other constituency test confirms its results.

In English, *and* can coordinate words or phrases of any category, while in other languages coordinators may work differently, for example with certain coordinators being limited to specific categories—for example, a language might have one coordinator for nouns, and another for verb phrases and sentences. But the same thing holds for all constituency tests; if you're looking at a new language, your first task will be to see if constituency tasks work the same way as they do in English, or if you can adapt tests so that they work in that language.

SUMMARY

Results of tests like these are how we investigate the structure of the mental grammar that underlies how people use the languages they know. We can't observe mental grammar directly, so observing how words behave is how we make inferences about how it must work. These four tests are tools that we have for observing how words behave in sentences. If we discover a string of words that passes these tests, then we know that the phrase is a constituent, and that tells us something about the organization of the sentence as a whole.

Not every constituent will pass every test, but if you've found that it passes two of the four tests, then you can be confident that the string is actually a constituent.

Check your understanding



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://ecampusontario.pressbooks.pub/essentialsoflinguistics2/?p=1483#h5p-133>

Navigation

If you are following the alternative path through this chapter that interleaves core concepts with tree structures, the previous section was [6.3 Structure within the sentence: Phrases, heads, and selection](#) and the next section is [6.5 Functional categories](#).

6.13 FROM CONSTITUENCY TO TREE DIAGRAMS

In this section we begin to introduce the formal notation of **tree diagrams** to represent the structure of phrases and sentences in a way that makes it easier to make specific and testable claims (hypotheses) about them.

Thinking back to [Section 6.1](#), one way of thinking about the goal of syntactic theory is that it's aiming to account for what languages users know about which sentences are grammatical, and which sentences are ungrammatical.

What constituents do we find inside sentences? Well, we know that a sentence consists of (at least) a subject and a predicate, that subjects are (usually) noun phrases, and predicates are often verb phrases. We might express this as a rule, known as a phrase structure rule.

(1) $S \rightarrow NP VP$

This rule says that wherever you have an *S* (a sentence), it is possible for that *S* to be made up of an *NP* (noun phrase) followed by a *VP* (a verb phrase).

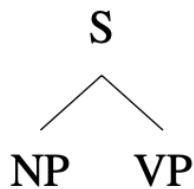


Figure 6.1
Schematic tree for
S containing an *NP*
subject and *VP*
predicate

Another way to represent the same idea is with a **tree diagram**, as in Figure 6.1.

Tree diagrams can express the same information as phrase structure rules, but can more efficiently express the output of multiple such rules; current syntactic theories are typically expressed in terms of constraints on possible trees, rather than in terms of constraints on phrase structure rules.

What kind of structure might we expect to find inside the *NP* subject? Here are some *NPs*—you might think of a sentence in which some or all of them can occur (remember that you can tell if a string in a sentence is a single *NP* by using a [replacement test](#) to try substituting a pronoun).

Phrase structure rules were central to the theory of syntax developed in Chomsky 1957, which kickstarted the modern field of generative syntax.

- (2)
- | | | |
|----|----------------------------------|------------------|
| a. | robots | (N) |
| b. | some robots | (D N) |
| c. | six robots | (Num N) |
| d. | the six robots | (D Num N) |
| e. | the six small robots | (D Num Adj N) |
| f. | robots from Boston | (N PP) |
| g. | the robots from Boston | (D N PP) |
| h. | the six small robots from Boston | (D Num Adj N PP) |

We could abstract across all of these, and write a general phrase structure rule for NPs, putting parentheses around all optional elements. A subscript “n” indicates that that element can be repeated any number of times.

- (3) $NP \rightarrow (Det) (Num) (Adj)_n N (PP)_n$

This can be read as:

An NP can consist of a determiner (optional), followed by a numeral (optional), followed by any number of adjectives (all optional), followed by a noun (required), followed by any number of prepositional phrases (all optional).

We could represent the structure of some of the specific NPs in (2) as in Figure 6.2. Each of these tree diagrams is consistent with the phrase structure rule in (3). The trees represent the pieces that an NP can be made up of: in Figure 6.2 we see a single N (*robots*) on the right, a determiner (Det) followed by an N (*the robots*) in the middle, and a determiner followed by a numeral followed by an adjective followed by a noun followed by a prepositional phrase (*the six small robots from Boston*) on the right.

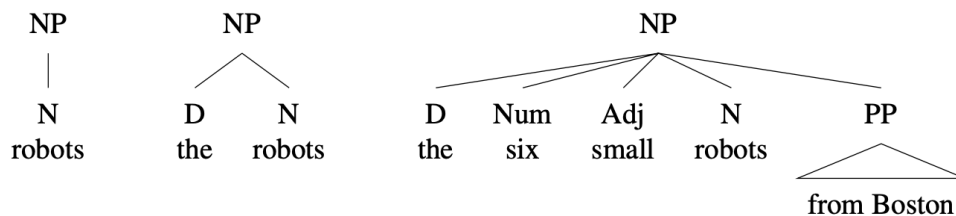


Figure 6.2 Tree diagrams for [*robots*], [*the robots*], and [*the six small robots from Boston*]

The phrase structure rule for NPs referred to prepositional phrases (PP). These phrases have fewer possible shapes than NPs do:

- (4) a. from Boston (P NP)
 b. outside (P)
 c. just outside (Deg P)
 d. way beyond my knowledge (Deg P NP)

Just like we did in (3) for NPs, we can abstract the possibilities for PPs listed in (4) into the single phrase structure rule shown in (5)

- (5) $PP \rightarrow (Deg) P (NP)$

We could then expand the “NP” symbol using our phrase structure rule for NPs above. That NP might contain another PP inside it—here we’ve encountered **recursion** again. Figure 6.3 shows tree structures for some of the PPs in (4).

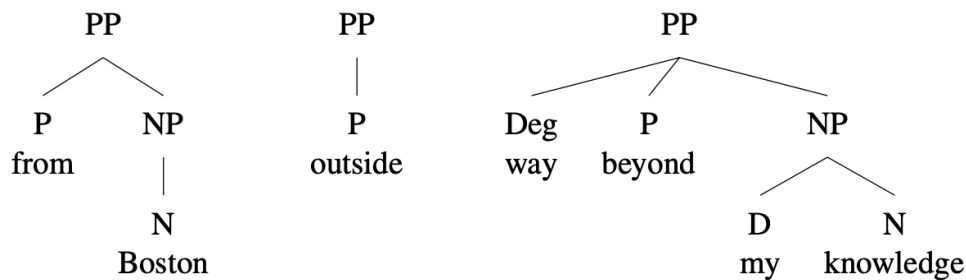


Figure 6.3 Tree diagrams for [*from Boston*], [*outside*], and [*way beyond my knowledge*]

Now let’s look at some verb phrases (VPs). In the following examples, the VPs are all in [square brackets].

- (6) a. The crew [repaired the ship].
 b. The captain [gave the crew orders].
 c. The spaceship [arrived].
 d. The crew [travelled across the galaxy].

How do we know these are VPs? Well, they come after the *subject* of the sentence (an NP in all these examples). In one case the predicate is a single word *arrived*—this word is a verb, so the only thing it could be is a verb phrase. All the other sequences in square brackets could be swapped into the same position as *arrived*, so they must be phrases of the same type. For example, For example, if we replace the predicate *arrived* in (6c) with [*gave the crew orders*], we get another grammatical sentence, shown in (7):

- (7) The spaceship [gave the crew orders].

This sentence might be a bit pragmatically odd—unless you assume the spaceship is an artificial intelligence that can give orders—but it is grammatical. Another test would be the **replacement test** for VPs, which involves replacement with *do* (you can review the replacement test in [6.4 Identifying phrases: Constituency tests](#)).

Based on these tests, we know that a verb by itself (like *arrived*) can be a VP, and that the object is inside the VP with the preceding verb. In (6) we have sentences with an intransitive VP containing just a verb, transitive VPs with verbs followed by a complement NP or PP, and a ditransitive VP that contains two object NPs.

Many ditransitive verbs in English can also appear with an NP and a PP (and some ditransitive verbs, like *put*, only allow the NP PP version). The alternation between a ditransitive [NP NP] as in (7) and [NP PP] as in (8) is called the **dative alternation**.

- (8) The spaceship [gave orders to the crew].

The VPs we've seen in this section can be derived with the following phrase structure rules:

- (9) a. $VP \rightarrow V_{\text{intrans}}$
 b. $VP \rightarrow V_{\text{trans}} NP$
 c. $VP \rightarrow V_{\text{ditrans}} NP NP$
 d. $VP \rightarrow V_{\text{ditrans}} NP PP$
 e. $VP \rightarrow V_{\text{trans}} PP$

If we looked at a wider range of VPs we'd also find that adverb phrases can go at the *beginning* or *end* of VPs, though not typically in the middle. So we'd end up with the following general phrase structure rule for VP:

- (10) $VP \rightarrow (AdvP) V (NP) (NP/PP) (AdvP)_n$

By putting together the structures we've proposed for NPs, PPs, and VPs, we're now in a position to show some trees for full sentences. Figure 6.4 shows a tree for the sentence [*The crew from Mars repaired the spaceship*].

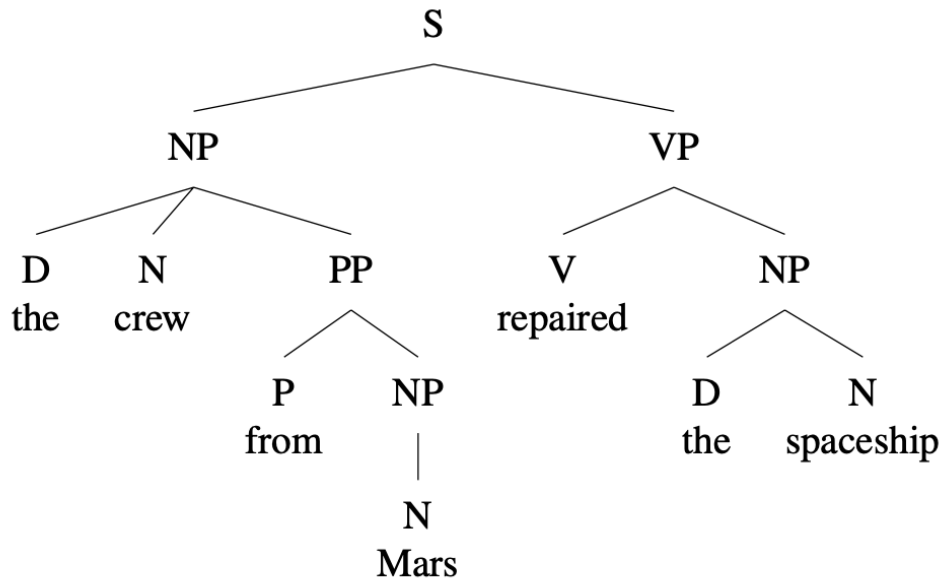


Figure 6.4 Tree diagram for *The crew from Mars repaired the spaceship.*

There are also some cases where a verb can be followed by an adjective phrase (*We are happy.*; *They seem nice.*). There isn't an easy way to collapse this with our previous phrase structure rule, with the notation we've introduced so far, so to account for this we could add a second phrase structure rule for VPs ($VP \rightarrow V \text{ AdjP}$).

We can also formulate phrase structure rules for various modifier phrases: AdvP, AdjP, and NumP.

Adverb phrases consist of an adverb (11a), preceded by an optional degree phrase (11b). You can also sometimes get a PP after an adverb (11c).

- (11) a. quickly
 b. very quickly
 c. quite quickly for a sloth

This is summarized by the phrase structure rule in (12):

- (12) $\text{AdvP} \rightarrow (\text{DegP}) \text{Adv} (\text{PP})$

Adjective phrases are similar:

- (13) The robot is [_{AdjP} very proud of itself].

(14) $\text{AdjP} \rightarrow (\text{DegP}) \text{Adj} (\text{PP})$

Number phrases are also modified by degree phrases!

- (15) a. exactly six
b. approximately 30
c. very many

(16) $\text{NumP} \rightarrow (\text{DegP}) \text{Num}$

The “objects” of adjectives are almost always expressed by PPs—that is, if there’s something in an AdjP that comes after the adjective, it usually can’t be an NP (or a VP), but instead has to be a PP.

Adjectives with NP complements

In English there is a very tiny set of exceptions to the generalization that complements of adjectives are PPs, though the exact number is different for different speakers. The one exception that all (or almost all) English speakers have is the adjective *worth*. So we can say:

(17) This object is [_{AdjP} worth [_{NP} a lot of money].]

Here the adjective is followed by the NP [*a lot of money*]. The NP has to be something that expresses a value.

Some English speakers, including most Canadian English speakers and some speakers in upstate New York and Pennsylvania, have another exception with the deverbal adjectives *finished* and *done*:

- (18) a. I am [finished my coffee].
b. The children are [done their homework].

If you aren’t from one of those places, you might need to use the verbal perfect (*have finished my coffee*), or use the preposition *with* (*are finished with their homework*), to express something similar to what’s expressed in (18a–b).

Some people allow a few more adjectives in this construction, but they’re all deverbal (that is:

derived from verbs). For example, people might allow NP complements with the adjectives *started* and *completed*. For more about this construction, see the [Done My Homework](#) entry at the [Yale Grammatical Diversity Project](#).

Let's return to phrase structure rules for whole sentences. We've already seen that sentences can consist of an NP followed by a VP:

(19) [The robot] [repaired the spaceship].

They can also have an **auxiliary** (modal or non-modal) as in (20a), and can have adverbs at the beginning or end as in (20b).

- (20) a. [The robot] will [repair the spaceship].
 b. Maybe [the robot] will [repair the spaceship] tomorrow.

You can also put **negation** in a sentence, as in (21)—though negation always requires an auxiliary in English (if there isn't already an auxiliary, we apply Do-support):

- (21) a. [The robot] didn't [repair the spaceship].
 b. [The robot] hasn't [repaired the spaceship].

From these we can get a full phrase structure rule with several optional elements, but an obligatory NP subject and an obligatory VP predicate:

(22) $S \rightarrow (\text{AdvP}) \text{NP} (\text{Aux}) (\text{Neg}) \text{VP} (\text{AdvP})$

Phrase structure rules are useful for describing the sequences that can occur in phrases of different types, but neither these rules nor the trees we've seen in this section do more than list the elements that can occur in phrases of different types.

In the remaining sections of this chapter, we'll explore a theory that limits possible tree structures to a few basic configurations, with the goal of explaining not only how languages can vary, but also explaining limits on the variation seen across human languages.

Check your understanding

Coming soon!

Navigation

If you are following the alternative path through this chapter that interleaves core concepts with tree structures, the previous section was [6.5 Functional categories](#) and the next section is [6.14 Trees: Introducing X-bar theory](#).

If your instructor has assigned Appendix 1 instead of the second half of this chapter, you should move on to [Appendix 1](#) now.

References

Chomsky, Noam. 1957. *Syntactic Structures*. The Hague: Mouton.

Wood, Jim. 2014. Done my homework. *Yale Grammatical Diversity Project: English in North America*. (Available online at <http://ygdp.yale.edu/phenomena/done-my-homework>. Accessed on 2024-09-12). Updated by Tom McCoy (2015), Katie Martin (2018), and Oliver Shoulson (2020).

6.14 TREES: INTRODUCING X-BAR THEORY

Constituency tests and phrase structure rules provide a useful starting point for thinking about the structure of possible sentences, but they don't really start explaining *why* certain structures are grammatical, or predicting what possible and impossible grammars might look like. In this section we introduce **X-bar theory**, which aims to make stronger predictions by restricting the shape of possible trees. It's called that because it introduces an extra layer of structure inside phrases called the "bar level".

To see why we might want to constrain what trees are possible, let's begin by thinking about a type of structure that's really easy to describe using a phrase structure rule:

- **Weird phrase structure rule:** $NP \rightarrow V (Adj) PP$

This rule is weird because it's a noun phrase that's missing the noun: we already saw in [Section 6.3](#) is that what makes something a noun phrase is precisely that it has a noun inside it. The restriction that all natural languages phrases have heads of the same category is the first limit we'll put on possible structures in X-bar theory:

- Every phrase (XP) has a head of the same category (X)

And this goes the other way as well: all **heads** (words) **project** (or "occur inside") a phrase of their category:

- Every head (X) projects a phrase of the same category (XP)

What this means is that even when a noun or verb—or any other category—doesn't obviously have any other words in the same phrase as it, it's still part of a phrase, an NP or a VP, just a one-word phrase. In other words, while the two sentences in (1) are in one sense very different (one has two words, the other has 11), in another sense they have the same structure: both sentences consist of an NP followed by a VP.

- (1) a. Cats sleep.
b. The many very fast spaceships carried a lot of valuable cargo.

By default, in X-bar theory we assume that the same constraints apply to all categories and phrases, and that they apply in all languages. In the absence of evidence to the contrary, we assume that determiners occur inside determiner phrases (DPs), degree words occur inside degree phrases (DegPs), and so on.

CHAPTER 7: SEMANTICS

This chapter is about linguistic meaning, particularly **semantics**: how the meaning of words combine to form the meaning of sentences. We will start by examining lexical meaning: what goes into the meaning of a word and other smaller linguistic expressions stored in your mental lexicon. We will examine various theories of lexical meaning and evaluate the pros and cons of each one. The latter half of the chapter focuses on case studies of linguistic meaning across categories and across languages, and along the way, we will think about what it means for the meaning of one word to combine with the meaning of another word. We will examine various data across categories and across languages in order to appreciate the complexity of human semantic competence.

When you've completed this chapter, you'll be able to:

- Acknowledge the plurality of linguistic theories concerning the status of the lexicon, explain the differences between each theory, and evaluate the pros and cons of each theory;
- Explain why the dictionary is not the ultimate authority of linguistic meaning;
- Analyse linguistic meaning critically based on descriptive observations;
- Gain a general understanding of what kinds of concepts lexical meaning encodes in language;
- Explain the difference between sense and denotation;
- Use diagnostics to identify entailments, implicatures, and presuppositions;
- Evaluate the usefulness of each kind of meaning in linguistic analysis;
- Appreciate the complexity and diversity of linguistic meaning.

7.1 LINGUISTIC MEANING



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We use the word “meaning” in various ways in our everyday lives. Consider (1)-(4).

- (1) In Japanese culture, what does it **mean** when a tea stalk floats vertically in your green tea?

takai mishin-o katte-mo tsukaikata-ga wakaranakereba **imi**-ga nai

- (2) expensive sewing.maching-ACC buy-even.if how.to.use-NOM understand.NEG.if **meaning**-NOM NEG.exist

‘There is no point in buying an expensive sewing machine if you don’t know how to use it’ (Japanese)

- (3) I said coffee is just as tasty as tea, but I didn’t **mean** it.

- (4) *Ode’imin* and *strawberry* **mean** the same thing.

(Examples inspired by Bach 1989)

In this chapter (Chapter 7) and the next chapter (Chapter 8), we will explore **linguistic meaning**: the sorts of things we have to know as language users when we produce and comprehend meaning. This excludes notions of “meaning” like in (1). (1) is not an example of *linguistic* meaning because the floating tea stalk (the object depicted in Figure 7.1) is not a linguistic expression. Sometimes we use the word *mean* or *meaning* to talk about our interpretation of the world. In this case, the vertical tea stalk “means” — or is symbolic of — good luck! Although interesting, this is not the kind of “meaning” we will focus on in this textbook.



Figure 7.1. A tea stalk floating vertically in a cup of green tea.

(2) also describes a kind of “meaning” that isn’t linguistic. Literally, the Japanese sentence in (2) means ‘There is no **meaning** in buying an expensive sewing machine if you don’t know how to use it’. This use of *imi*, or *meaning*, is about the purpose of some action. Here, we are saying that there is a relation between an action and some sort of ideal end result that the act is supposed to lead to. This, again, is not the kind of “meaning” we are interested in in this textbook.

The notion of **meaning** in (3) is certainly about language, in the sense that we are evaluating what someone said. “Meaning” here seems to be something about sincerity: (3) is roughly paraphrasable as ‘I said coffee is just as tasty as tea, but I wasn’t being sincere when I said this’. This is not the kind of “meaning” we will be concerned with in this particular chapter, but Chapter 8 (Pragmatics) will discuss some points that will be relevant to this notion of “meaning”.



Figure 7.2. A strawberry.

“Meaning” as used in (4) best exemplifies what we mean by *meaning* (= linguistic meaning) in this chapter: the words *ode’imin* and *strawberry* **mean** the same thing. *ode’imin* is an Ojibwe (Anishinaabemowin) word, and *strawberry* is an English word. But what does it mean for them to “mean the same thing”? What we want to figure out in this chapter is something like this: what is the knowledge about the word *ode’imin* that an Ojibwe speaker stores in their head, such that when they utter the word, they know it points to that juicy red fruit with small seeds on the surface (Figure 2)? Similarly, what is the knowledge about the word *strawberry* that English speakers have in their head, such that they know that this word in English points to this same fruit? Both the word *ode’imin* and the word *strawberry* carry some sort of content, and this information allows for Ojibwe and English

language users to say “Yes, that is an *ode’imin/strawberry*” or “No, that is not an *ode’imin/strawberry*” for any given object. In this way, the two words “mean the same thing”.

Did you know? *Ode’imin* in Ojibwe literally means ‘**heart berry**’ (in reference to its heart-like shape; see Figure 7.2)! The origin of the English version *strawberry* is not clear according to the Oxford English Dictionary. One explanation is that the name is perhaps in reference to the “straw”-like particles on the fruit’s surface.

Note that when we say “the meaning of *ode’imin*“, we are talking about the meaning of the **linguistic expression** *ode’imin*, not the actual fruit in real life. In this chapter, we are asking what meaning is attached to the stream of sounds *ode’imin* (IPA: [odeimin]) in Ojibwe, not to the actual fruit like in Figure 7.2. Similarly, we are asking what meaning is attached to the stream of sounds *strawberry* (IPA: [stɹɔbɛɹi]), again, not to the actual fruit. For signed languages, we want to know what meaning is attached to a certain sign: like what is shown in Figure 7.3, which is the sign for ‘strawberry’ in American Sign Language. **Linguistic expressions** are the words, phrases, and sentences we produce as language users. Following writing conventions in linguistics, linguistic expressions that are [mentioned](#) in paragraphs will be italicized in this textbook. So when we write “*strawberry*” (in italics), we are referring to the linguistic expression (the word) that points to the object in Figure 7.2. When we write “strawberry” (no italics), we are actually talking about the object in Figure 7.2, the actual fruit. So “**the strawberry** has 4 syllables” makes sense, but “**the strawberry** has 4 syllables” doesn’t (because a fruit doesn’t have syllables).



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Figure 7.3. Three variations of the word ‘strawberry’ in American Sign Language.

The English word **semantics** comes from the Ancient Greek word that means ‘significant’ (*sēmantikós*). This comes from a word that means ‘to show, signify’ (*sēmaínein*).

Semantics is the study of linguistic meaning: how the meaning of words combine to form the meaning of sentences. In this chapter, we will analyse linguistic meaning from various perspectives. One of the big conclusions of this chapter (Chapter 7) and the next chapter (Chapter 8, Pragmatics) is that there are many different kinds of linguistic meaning. Each way of thinking about meaning gives us important insight into how we produce and understand meaning as language users. We will start this chapter by looking at what people probably usually think of when they hear “meaning”: the **sense** of words that we store in our heads. The sense of *strawberry* would be whatever “meaning” of that word English users store in their head. After our discussion of sense, we will look at meaning in terms of the **denotation** of linguistic expressions: what linguistic expressions point to in the real world. As we’ve seen already, one way of thinking about how *odé’imin* and *strawberry* “mean the same thing” is to say that these words both point to (or **denote**) the same fruit in the actual world. As we will see, thinking about meaning in terms of the linguistic expression’s denotation also helps us explain semantic patterns in language.

The idea that there are two ways of thinking about meaning — sense and denotation — was originally proposed by German philosopher, logician, and mathematician **Gottlob Frege** in 1892.

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7.2 COMPOSITIONALITY: WHY NOT JUST SYNTAX?



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Consider the following sentence.

- (1) The raccoon clothing store was doing a photo contest so I submitted a picture of a Toronto raccoon wearing a bright pink bandana with lime green polka dots on it.

This is likely a sentence you have never heard before in your life, but you are still able to comprehend it and understand what it means. The reason for this is because you have the meaning of each morpheme that appears in this sentence stored in your head, and by combining those meanings, you are able to get the meaning of the overall sentence. This is the principle of **compositionality**: the meaning of a complex linguistic unit results from the individual meanings of its subparts, and how these subparts were combined. This means that the meaning of a sentence is determined by the meaning of the words that it contains, and how these words were put together. It also means that the meaning of a multi-morphemic word is determined by the meaning of the morphemes it contains, and how they were put together. So for example, the meaning of the word *unhelpfulness* is the combination of the meaning of *un-*, *help*, *-ful*, and *-ness*, put together as [[un- [help -ful]] -ness].

Compositionality is not just a matter of *what* linguistic pieces you put together: it also matters *how* you put the pieces together, too. This is relevant to the notion of **constituency**, or unithood, from Chapter 6 (Syntax). For example, per compositionality, the meaning of the sentence *That fluffy cat will chase the raccoon* is the combination of the meaning of the words *that*, *fluffy*, *cat*, *will*, *chase*, *the*, and *raccoon*. What Chapter 6 taught us is that you don't just string these words together like a necklace to form a sentence: the composition of a sentence happens in a more systematic way, with smaller substructures within. For example, *that fluffy cat* and *the raccoon* are syntactic units, or **constituents** in this sentence. *That fluffy cat will* is not a constituent in this sentence, but *chase the raccoon* is. Following this syntax (see Figure 7.3), we can analyse that word meanings combine in the same structured way, too. The meaning of *the* and *raccoon* will combine first, then the meaning of *the raccoon* will combine with the meaning of *chase*, and so on. Keep this in mind as you read this chapter. If you need a refresher on syntax, you may want to reread Chapter 6 (Syntax) before proceeding with the rest of this chapter. This chapter will assume that you have completed Chapter 6 (Syntax) already.

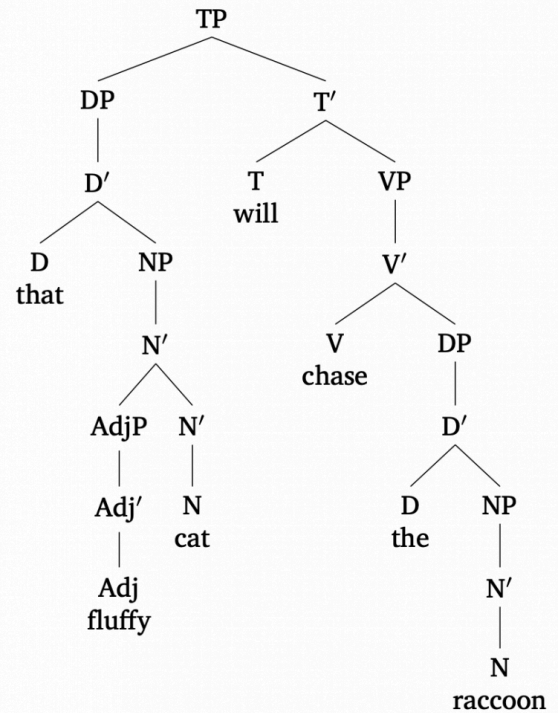


Figure 7.3. Syntactic structure for *That fluffy cat will chase the raccoon*.

We just said that compositionality is about how linguistic pieces are put together. Since we talked about this notion in detail in Chapter 6 (Syntax), you may be wondering how this chapter on semantics is different. To a large extent, syntax and semantics go hand-in-hand. If two words go together as a syntactic unit, it's fair to hypothesise that they form a semantic unit, too, and vice versa. For example, in our sentence *That fluffy cat will chase the raccoon*, *fluffy* and *cat* form a syntactic unit: it's a noun phrase. *Fluffy cat* is also a semantic unit: the meaning of *fluffy* gives us more information about what kind of *cat* we are talking about. So then, why do we need to talk about semantic composition when we've already talked about syntactic composition? Put another way, we know that an adjective and a noun form a unit (NP), a determiner and a noun phrase form a unit (DP), a verb and a determiner phrase can form a unit (VP), and so on — so what more is there to say about compositionality? If we know the syntax, doesn't the semantics naturally follow?

The answer is “no, not necessarily.” We see places in language where the syntactic categorisation of a linguistic expression does not predict everything about the semantics of it. Consider the determiner phrases (DPs) *a scientist* in (2) and (3), for example.

- (2) Vera greeted **a scientist**.
 (3) Vera became **a scientist**.

Syntactically, *a scientist* is a DP in both (2) and (3). However, just because they have the same syntactic form does not mean that their semantics is the same: *a scientist* in (2) does not mean the same thing as *a scientist* in (3). The meaning of *a scientist* in (2) is **referential**: it refers to some existing scientist. If this scientist is named Gladys, (2) is equivalent in meaning to *Vera greeted Gladys*. *A scientist* in (3) is not referential; this time, it is not pointing to some existing scientist. *A scientist* in (3) is **predicative**: it expresses a property about the subject, which in this case is *Vera*. Adjectives are often predicative and can be seen in a construction like (3), too: e.g., *Vera became cheerful*. In English, sentences with a referential DP can be **passivized** as in (4), but sentences with a predicative DP cannot be passivised, as (5) shows (Williams 1980, 1983).

- (4) **A scientist** was greeted by Vera.
 (5) * **A scientist** was become by Vera.

The point of these examples is that we might expect (5) to be grammatical based on the syntax of *a scientist*: it's still a DP, after all. However, the reality is that (5) is ungrammatical. Clearly, this ungrammaticality is not due to the syntax of *a scientist*: we know that some DPs can be the subject of a passive construction, as seen in (4). Rather, it is the *meaning* of the DP that is causing the ungrammaticality in (5): DPs that are predicative in their meaning cannot be passivised. This is one example of how the meaning of a linguistic expression really matters in language, not just the syntax.

There are also places in language where the reverse is true: we might expect something to be grammatical based on the semantics of the expression, but it winds up being ungrammatical due to the syntax. Consider the prepositions *before* and *during* in (6) and (7), respectively.

- (6) Alan sneezed **before** { the meeting with his student / the race / the party / the event }.
 (7) Alan sneezed **during** { the meeting with his student / the race / the party / the event }.

Based on this data, the prepositions *before* and *during* seem to be semantically compatible with expressions that point to an event of some sort. For *before*, the syntactic type of this event does not seem to matter. For example, it can also take a full clause (TP/CP) as an object:

- (8) Alan sneezed **before** { he met with his student / he raced / he partied / he attended the event }.

During, however, is syntactically pickier. It does not allow for a clausal complement:

- (9) * Alan sneezed **during** { he met with his student / he raced / he partied / he attended the event }.

The DP *the meeting with his student* and the sentence *he met with his student* semantically both point to the event of Alan meeting with his student, but syntactically they are different types of units. (9) is an instance in language where just the semantics is not sufficient for explaining the descriptive patterns; the syntax clearly matters too.

To summarise, syntax certainly is important when thinking about how sentences are composed, but it's also important for us to examine the semantics separately, too.

Check your understanding



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